

2. Convert X and y from list types to numpy array to feed it into our model using the numpy library.
3. Finally, plot the above data to verify the number of different labels.

After checking for the images and labels of data, now we have to check for the shape of data. It can be checked using the `.shape` argument. We find that our image is a tensor of rank 4 or a four dimensional array with dimensions 4000 x 150 x 150 x 3. They correspond to the batch size, height, width and channels respectively.

Splitting our dataset

Now we have to perform an important step, which is to split our data into train and validation set. This is one of the most important things to do before training a model. For splitting, we are going to use a popular function called `sklearn` from the multipurpose python library `scikit-learn`.

1. Import `train_test_split` from `sklearn`.
2. Give inputs to the function of the percentage of data to be assigned to the validation set and the train set. A usual practise is 20% for validation set and rest 80% for the training set.
3. Print the shape of the new train and validation set to verify the splitting of data.
4. Specify the batch size that tells the number of images that will be fed into the training model all at once. It should be in multiples of two; for instance, we have used a batch size of 32 in our model.

With this, we have completed the pre-processing of data for the model.

Keras Modules Import

We need to import keras layers, models and optimiser modules.

Keras layers: It contains different types of layers that are used in deep learning such as:

1. Convolutional layer: It is mostly used in computer vision. The primary purpose of the convolution layer is to extract features from the input image. This preserves the spatial relationship between pixels by learning image features using small squares of input data. Finally, the obtained matrix is known as the feature map.
2. Pooling layer: Also called subsampling or downsampling layer, it is also used in computer vision. Pooling reduces the dimensionality of each feature map but retains the most important information. There are many types of pooling layers such as Max Pooling and average pooling.